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purify: An R package for resampling and stratified data

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Software

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Summary

`purify` is an R package designed for resampling and testing of data, aimed at researchers and practitioners who analyze complex datasets, often with unbalanced strata, multiple predictors, or a dependent output variable. This package enables users to perform robust statistical analyses using test statistics and resampling of variables with and without relation to the output and other stratification variables. The resampling analysis can be conducted on statistical summaries (e.g. mean square error); coefficient, forecast, and model statistic estimates; and other custom defined statistics, `purify` offers versatile resampling settings, including block, sliding window, and stratified resampling with and without replacement. These methods are also extended to cross-validation and confidence intervals. Each method can be tailored to specific data structures and research questions. With its intuitive interface and customizable options, `purify` streamlines the process of hypothesis testing and estimation of statistical significance.

Statement of need

Unbalanced data are widely observed, yet methods for analysis can be unwieldy, overly complex, or missing implementations. `purify` fills this gap by providing and organizing statistical tests robust to various assumptions and an extensive collection of resampling methodology. These tests include normality, anova, and two-sample tests along with statistics to quantify stratification effects.

Resampling is a fundamental technique in estimating the distribution of statistics, testing hypotheses, and deriving confidence intervals–especially when analytical solutions are impractical. When assumptions for statistical tests are under question, resampling provides another tool to assess their effectiveness. Many R packages provide basic resampling methods but lack specialized support for complex structures. Such structures may contain dependence and variables which should not be resampled. Analysis of these structures can be unwieldy to investigate in other packages. `purify` offers a flexible framework to enable data scientists and researchers to perform and compare targeted, customizable resampling schemes that account for data structure. The methods are compatible for user-defined functions and outside models, making `purify` useful for rigorous hypothesis testing and model evaluation.

Permutation tests can be computationally intensive, so speed is considered throughout `purify`. Cross-validation, forecasting, visualization, and summary functions extend the framework and support interpretation.

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State of the field

Several R packages provide important components of resampling and statistical analysis. The `boot` package provides a general framework for bootstrap methods (Canty & Ripley, 2024), while `rsample` supports resampling objects for modeling workflows (Kuhn & Wickham, 2024). Packages such as `caret` provide cross-validation and model evaluation tools (Kuhn, 2008), and packages such as `car` and `PMCMRplus` provide tests for variance heterogeneity and group differences (Fox & Weisberg, 2024; Pohlert, 2024). These packages are valuable, but users often need to combine several interfaces when analyzing structured, dependent, or unbalanced data.

`purify` centralizes and expands resampling capabilities within a common framework. No single cited package combines structured block and stratified resampling, unequal-group handling, resampling-based statistical tests, forecasting intervals, and user-defined analysis functions in one interoperable interface. Its contribution is therefore not simply to collect existing tests, but to provide workflows for structured resampling. In particular, `purify` supports separate and sliding-window block resampling, stratified resampling with unequal group sizes, resampling with or without replacement, and user-defined functions to structured data. These features are implemented through functions such as `resample()` and `observation_probability()`. The package also extends this framework to simulation-based hypothesis testing, pairwise group comparisons, variance estimation, effect-size estimation, forecast confidence intervals, and dependent cross-validation. For example, `resample_welch_anova()` uses resampling to evaluate an ANOVA statistic under a simulated null distribution, while `confidence_intervals()` supports bootstrap forecasting with user-supplied prediction functions and rolling training windows. `resample_differences()` and `resample_variance()` provide group-specific bootstrap estimates and intervals using the same underlying resampling framework.

Some procedures included in `purify`, such as Games–Howell comparisons, normality tests, and multinomial calculations, are established methods rather than newly invented statistical procedures. Their inclusion provides a consistent interface and allows them to be used alongside the package's structured resampling and simulation methods.

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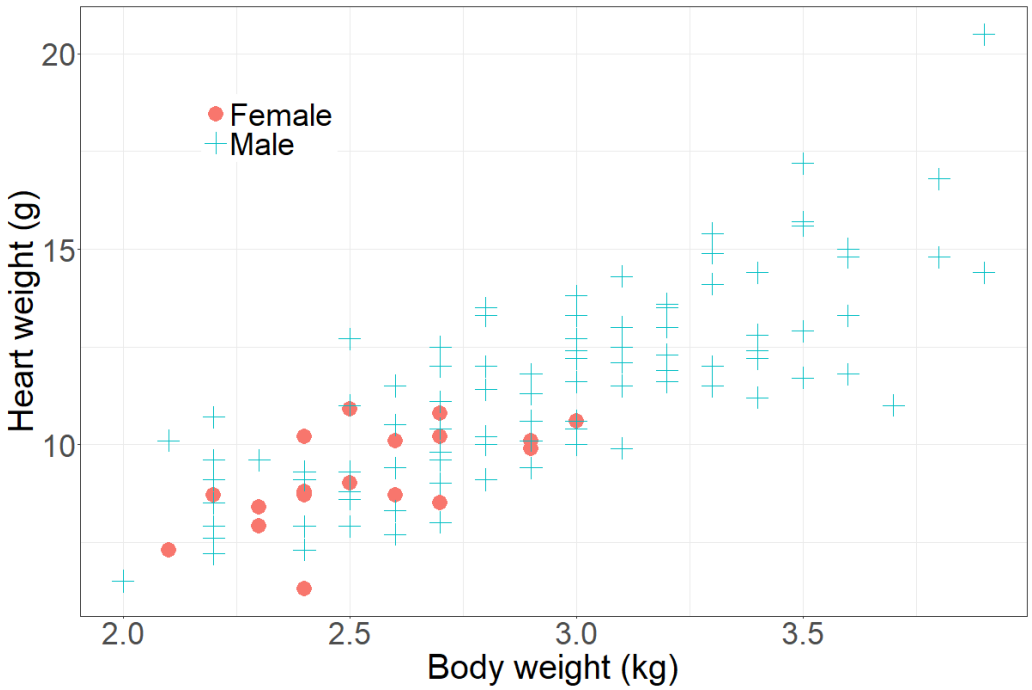


Figure 1: Subcats. Body and heart weights of cats with respect to their sex.

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Package functionality

Assessing whether the data is homogeneous in its variance, exhibits normality, or has significant differences between strata often requires extensive testing. `purify` offers functions such as `normality_tests()`, `variance_tests()`, and `group_tests()` to investigate several test statistics at once. Resampled versions of statistics are provided to determine estimates and confidence intervals with fewer assumptions, e.g. see `resample_variance()`.

A primary function in `purify` is `resample()`. This function offers clear input parameters to simplify the process of selecting or evaluating the correct methodology, even for user-defined functions. The multistep selection allows for combinations of dependent data, unbalanced data, and resampling to be performed with and without replacement. The flexibility enables users to adapt `purify` to diverse data contexts and hypothesis-testing requirements. See also `cross_validation()` and `confidence_intervals()`.

Software design

The package is organized around `resample()`, which provides a common engine for ordinary, block, sliding-window, and stratified resampling. Rather than creating separate functions for every combination of sampling assumptions, the design exposes these choices as parameters. This reduces duplicated code and allows the same resampling logic to be reused for bootstrap estimation, permutation-style procedures, simulation-based tests, and model evaluation.

The `fn` argument allows users to apply an arbitrary function to each resampled data set. This supports summaries, fitted models, forecasts, and user-defined test statistics without requiring the package to anticipate every analysis. Arguments for strata, block size, replacement, sample sizes, and columns that should not be resampled provide control over common sources of dependence and imbalance.

Higher-level functions build on these design choices rather than implementing independent

resampling systems. For example, the resampling functions for Welch ANOVA, pairwise differences, and variance estimates use related data and simulation structures, while the forecasting and cross-validation functions adapt them to sequential data. This design prioritizes flexibility and reuse, with the trade-off that users must understand their data structure and choose sampling parameters appropriate to their analysis.

These choices involve deliberate trade-offs. A parameterized common engine reduces duplicated implementations and keeps resampling behavior consistent, but it gives users more responsibility for selecting valid sampling settings. Allowing arbitrary functions makes the framework useful for models, forecasts, and custom statistics, while also requiring those functions to be appropriate for each resampled data set. Block and sliding-window methods better preserve dependence in sequential data, but can reduce the effective sample size and increase computation.

Research impact statement

`purify` has been developed in response to research analyses involving unevenly sized groups, stratified data, dependence, and resampling-based model evaluation. It was used to conduct the statistical analyses of study data in projects reported by Tetui et al. (2022) and Alexander et al. (2024), focused on healthcare and citations. These applications motivated the package's emphasis on structured resampling and user-defined analysis functions.

The repository also provides reproducible examples, package tests, datasets, vignettes, and documentation that demonstrate the methods on applied data. Together, these materials provide a starting point for researchers who need to adapt resampling and simulation procedures to their own data structures. The package is designed to support further research use by allowing new summaries, models, and test statistics to be supplied through user-defined functions without modifying the resampling engine.

Example

`purify` provides in-depth articles on the package website. For example,

- The *purify* article describes the core features of `purify` and includes simulations and real data examples to demonstrate the functions.
- The *cats* article details investigation on real data.

We consider a subset of the cats (*subcats*) data set below. We use sex and body weight to estimate heart weight; see Figure 1. Similar to many real-world examples, the data is highly imbalanced. Nonetheless, sex and body weight are both useful in understanding heart weight. In particular, female cats have lower body weights and have a lower heart weight even for the same body weight when compared to male cats.

Table 1: Subcats models. Models of cats using body weight and sex to predict heart weight.

	Intercept 95% Conf Int	Sex (M) 95% Conf Int	Body weight 95% Conf Int	MSE
Single model	-1.486 (-3.236, 0.264)	0.617 (-0.139, 1.372)	4.208 (3.573, 4.843)	2.258
Resampled model	-1.427 (-3.603, 0.610)	0.620 (0.051, 1.167)	4.186 (3.425, 5.005)	2.192

Let the linear model be defined as heart weight predicted by an intercept, body weight, and sex. Estimates for the coefficients and the confidence intervals for each parameter are given in Table 1. When applying the linear model directly on the data, only body weights appear to

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significantly impact heart weight. For resampled data, where samples are taken to create more evenly sized groups based on sex, both sex and body weight are determined to be significant. The cost for this simple example is that the confidence interval on body weight is larger. While additional simulations or modifying the resampling scheme may mitigate such losses, it is important to consider such effects. Often prediction error, such as mean square error (MSE), is more important and in this case, the resampled model also performs better. See articles for information on other functions and additional analysis on this and other data.

AI usage disclosure

The software's core research goals, statistical methods, architecture, and initial implementation were developed by the human authors. During revision, OpenAI Codex was used to assist with code review, documentation improvements, and editorial drafting. AI assistance was used as a review aid rather than a substitute for author decisions about the methods or software design.

The authors reviewed, edited, and validated all AI-assisted suggestions, including by inspecting the source code, adding and running tests, checking package documentation, and running R package checks. The authors remain responsible for the accuracy, originality, licensing, and scientific content of the software and manuscript.

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Contributions to `purify` are welcome and notable recognition is given to all who raise awareness of deficiencies in the package via the GitHub repository.

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